

## COMPRESSED AIR TREATMENT

# Refrigerated Air Dryers

High-efficiency direct expansion compressed air dryers engineered for maximum moisture separation and continuous industrial duty.



**20 – 2000  
CFM**

CAPACITY RANGE

**+3°C**

PRESSURE DEW  
POINT

**R-134a / R-  
407C**

ECO REFRIGERANT

**16 Bar**

MAX WORKING  
PRESSURE

Industrial Catalog  
Series  
Edition 2026-27

## Engineering Excellence in Air Treatment

Win Equipments Refrigerated Air Dryers deliver reliable moisture extraction to safeguard downstream pneumatic machinery, instrumentation, and finished goods. Utilizing advanced 3-in-1 aluminium plate heat exchangers, our systems achieve optimal heat transfer with minimal pressure drop.

### ✓ 3-in-1 Compact Heat Exchanger

Combines air-to-air heat exchanger, evaporator, and moisture separator into a single brazed aluminium block, drastically cutting energy consumption.

### ✓ Stable +3°C Dew Point

Precision hot gas bypass valve maintains constant dew point control across fluctuating plant loads, preventing moisture condensation in air lines.

### ✓ Eco-Friendly Refrigerants

Charged with non-ozone depleting R-134a and R-407C refrigerants conforming to international environmental safety standards.

### ✓ Zero-Loss Electronic Drain

Microprocessor-controlled automatic condensate drain ensures condensate discharge without compressed air wastage.

### Thermodynamic Working Principle

Warm, moisture-laden compressed air enters the air-to-air precoolers where it is pre-cooled by outbound chilled air. It then passes into the refrigerant evaporator where cooling to +3°C causes moisture vapour to condense into liquid droplets. The high-efficiency integrated demister separates condensed water and oil, discharging them via the automatic drain valve. The dry air is reheated before exit, ensuring zero pipe sweating.



### Built for Rigorous Industrial Environments

Heavy-duty epoxy powder-coated steel casing resists corrosion in humid factory environments. Designed for ambient operating temperatures up to 50°C and inlet air temperatures up to 60°C.

## Technical Specifications & Performance Data

Standard operating conditions: Inlet air temperature 38°C, Ambient temperature 35°C, Working pressure 7.0 bar(g). Maximum working pressure: 16 bar(g).

| MODEL CODE | CAPACITY (CFM) | CAPACITY (M <sup>3</sup> /MIN) | POWER (KW) | REFRIGERANT | AIR CONN. (BSP) | DIMENSIONS (L×W×H MM) | WEIGHT (KG) |
|------------|----------------|--------------------------------|------------|-------------|-----------------|-----------------------|-------------|
| WRD-20     | 20             | 0.57                           | 0.35       | R-134a      | 1/2"            | 480 × 360 × 520       | 32          |
| WRD-40     | 40             | 1.13                           | 0.52       | R-134a      | 3/4"            | 520 × 400 × 580       | 41          |
| WRD-60     | 60             | 1.70                           | 0.68       | R-134a      | 3/4"            | 560 × 440 × 640       | 52          |
| WRD-100    | 100            | 2.83                           | 0.95       | R-134a      | 1"              | 620 × 490 × 710       | 68          |
| WRD-150    | 150            | 4.25                           | 1.30       | R-407C      | 1-1/2"          | 710 × 560 × 820       | 85          |
| WRD-200    | 200            | 5.66                           | 1.75       | R-407C      | 1-1/2"          | 780 × 610 × 890       | 105         |
| WRD-300    | 300            | 8.50                           | 2.40       | R-407C      | 2"              | 860 × 680 × 980       | 135         |
| WRD-500    | 500            | 14.16                          | 3.80       | R-407C      | 2" Flanged      | 1020 × 780 × 1140     | 190         |
| WRD-750    | 750            | 21.24                          | 5.50       | R-407C      | 2-1/2" Flanged  | 1180 × 890 × 1280     | 260         |
| WRD-1000   | 1000           | 28.32                          | 7.20       | R-407C      | 3" Flanged      | 1350 × 980 × 1420     | 340         |
| WRD-1500   | 1500           | 42.48                          | 10.50      | R-407C      | 4" Flanged      | 1580 × 1120 × 1600    | 480         |
| WRD-2000   | 2000           | 56.63                          | 14.00      | R-407C      | 5" Flanged      | 1780 × 1250 × 1750    | 620         |

**Note:** Higher capacity models up to 5,000 CFM and water-cooled condenser configurations are available upon request. Win Equipments reserves the right to modify technical data without prior notice in pursuit of continuous technological innovation.

## Key Industries & Applications

Trusted by leading manufacturing units across India for process air reliability and ISO 8573-1 air purity compliance.

### Automotive & Spray Painting

Prevents surface blemishes, cratering, and paint blistering by removing all condensable moisture.

### Pharmaceutical & Cleanrooms

Ensures dry carrier air for packaging lines, tablet coating, and pneumatic tablet transfer.

### Food & Beverage Packaging

Prevents biological growth and contamination in PET bottle blowing and canning systems.

### Textile & Spinning Mills

Protects high-speed spinning rotors, auto-coners, and loom nozzles from water droplets.

### Laser Cutting & CNC Machinery

Guarantees clean assist air for precision optical cutting heads and tool changers.

### Electronics & Semiconductor

Prevents micro-corrosion and static issues in automated pick-and-place robotics.

## WIN EQUIPMENTS

### Head Office & Manufacturing Works:

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**Quality Management:** ISO 9001:2015 Certified (IAF & DAC Accredited)



SCAN TO VIEW PRODUCT